Newlines are scope: Negation lost in clinical-trial text normalization

Eligibility criteria determine who may enter a clinical trial, but they are distributed as free text that must be converted into computable constraints. Negation is consequently central to clinical information extraction, from NegEx (Chapman et al. 2001) to Criteria2Query (Yuan et al. 2019; Fang et al. 2022) and recent trait-decomposition pipelines (Chen et al. 2026). These approaches model lexical cues, entities, and logical relations. We identify a prior representational problem: a criterion's line boundary can be its only overt scope boundary. ClinicalTrials.gov preserves Markdown-like bullets, but newline-flattening turns sibling criteria into one punctuation-delimited string. We ask how often this lossy normalization creates a possible negation span that contradicts the list structure.

Data and method. We queried the public ClinicalTrials.gov API on November 5, 2026 for the 250 most recently updated recruiting trials in each of four domains: cancer, heart disease, diabetes, and depression. After cross-query deduplication, the sample contained 989 trials; 970 had explicitly bulleted or numbered criteria. We identified 17,673 list units, joining unmarked wrapped lines to their preceding bullet before analysis. This avoids treating display wrapping as structure.

We located a conservative set of clinical negation cues: *no, not, without, never, neither, cannot, can't, must not, may not, absence of, free of*, and *negative for*. A BOUNDARY-SENSITIVE token is a cue in a list unit that (i) lacks final punctuation and (ii) is followed by another unit in the same inclusion or exclusion section. A line-aware representation closes scope at the list boundary. A punctuation-only baseline deletes the boundary and holds scope open until *.*, *!*, *?*, *;*, or *:*. This diagnostic measures a structural failure opportunity, not an observed error by a particular system.

Example (1), from NCT07318805, shows three sibling exclusion criteria. The direct object of *controlled* ends in (1a); neither (1b) nor (1c) is in the scope of *cannot*. After newline deletion, however, no punctuation separates the three predicates.

(1) a. *Hypertension that cannot be controlled by optimal medical therapy*
    b. *Renal impairment*
    c. *Hepatic dysfunction*

Results. Of 17,673 criteria, 2,084 (11.8%) contained a cue. Among cue-bearing units, 892 (42.8%) were boundary-sensitive (Table 1). With newlines removed, the punctuation baseline crossed a median of four subsequent criteria and 67 words before reaching a terminator. Rates differed by clinical domain, $\chi^2(3) = 28.05$, $p < .001$, Cramer's $V = .116$: oncology had the highest rate (47.7%) and psychiatry the lowest (32.5%). Oncology's longer spill spans (median five criteria and 120 words) show that the effect is not merely one adjacent short phrase. In a deterministic manual audit of 40 events (10 per domain), 36 (90%) were independent sibling constraints; four were coordinated or nested continuations. This motivates retaining indentation and overt continuation markers rather than treating every newline identically.

Table 1: Boundary-sensitive negation cues by domain. "Trials" includes only records with structured list units.

Domain	Trials	Units	Cue units	Boundary-sensitive
Oncology	241	5,417	975	465 (47.7%)
Cardiology	245	4,057	333	140 (42.0%)
Diabetes	240	3,957	354	150 (42.4%)
Psychiatry	244	4,242	422	137 (32.5%)
Total	970	17,673	2,084	892 (42.8%)

Analysis. In this register, typography supplies a constituent boundary. Eligibility criteria are telegraphic predicates whose subject (the potential participant) and modal force (must/must not satisfy) are inherited from the section heading. A same-level bullet packages one constraint under that shared frame. The line break is therefore not decorative whitespace: it closes the current constraint before the next predicate is evaluated. Flattening (1) erases the evidence that *renal impairment* is a separate exclusion, not an argument or continuation of *cannot be controlled*.

We propose BOUNDARY-AWARE SCOPE PRESERVATION. Before lexical or neural analysis, each same-level criterion boundary is converted to a typed sentinel, `<CRIT>`; ordinary wrapped lines remain inside the unit, while indented sub-bullets attach to their parent. Negation scope may not cross `<CRIT>` unless an overt continuation marker (e.g., *and* introducing a coordinated subcriterion) licenses attachment. This preprocessing change eliminates all 892 mechanical spill opportunities by construction while leaving within-unit negation to existing systems. It is compatible with rule-based scope systems, semantic parsers, and language-model tokenizers.

The finding complements annotated resources such as the 1,006-document Leaf Clinical Trials Corpus (Dobbins et al. 2022). Those resources richly represent entities and relations, but the input serialization itself remains part of the semantic problem. Recent work decomposes complex criteria into atomic traits and explicitly annotates negation (Chen et al. 2026); our result identifies a cheap, upstream boundary signal for that decomposition. It also offers a controlled benchmark: pair each original list with a newline-flattened version and test whether a parser preserves sibling polarity and logical independence.

Three predictions follow. First, boundary preservation should improve precision most in oncology, where negation cues and unpunctuated lists are most dense. Second, models trained on flattened corpora should be less robust to original list formatting than models trained with typed boundaries. Third, human judgments should assign narrower scope to the formatted version of minimal pairs like (1), even when lexical content is held constant. The current cue-based study is not a gold-standard error analysis; rather, it quantifies how frequently a commonplace normalization destroys a linguistically licensed scope boundary. The manual audit was small and single-reviewed; clinician annotation remains necessary. In a safety-relevant medical register, preserving the original boundary should nevertheless be the default.

References

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